Generated diffusion window: GPU kernels with native wall corrections

Diagnostic sections policy • channel-simple validated; channel candidate untested • automatic pricing unavailable

Both paths: SGS direct → native wall / viscosity work → stage initialization → advection, filtering and forces

Original numerical order and wall model retained; channel's active filtering stays in the native prefix.

Separate ownership: SGS publishes before this native prefix. The new window starts inside the selected diffusion call.

SGS-DIRECT CONTROL

Diffusion remains native

CPU U diffusion

then 12 native wall corrections

CPU V diffusion

then 12 native wall corrections

CPU W diffusion

then 12 native wall corrections

Original completion and barriers retained

No diffusion PCIe traffic in this control

SGS DIRECT + GENERATED DIFFUSION

One diffusion owner; one original OpenMP team; no owner boundary

Device: U / V / W + viscosity

Shared read-only fields across all three kernels

Host: U / V / W + viscosity

Current mirrors retained for native readers

Entry: upload missing input sections, including stars

Read-only host mirrors need no download

1 GPU U diffusion

Read U, viscosity and U*; update U*

Publish dirty U* sections

2 CPU wall corrections

12 original worksharing loops update U*

3 GPU V diffusion

Read V, viscosity and V*; update V*

Publish dirty V* sections

4 CPU wall corrections

12 original worksharing loops update V*

5 GPU W diffusion

Read W, viscosity and W*; update W*

Publish dirty W* sections

6 CPU wall corrections

12 original worksharing loops update W*

Close: publish remaining dirty stars; preserve sections already corrected on the host

Native accumulation and the remaining RK step stay outside the window

Stars remain live after return. This checkpoint does not retain ownership through accumulation or across stages.

Original operation order

Publication obligations; observed totals in transfer plot

U* = Ustar

Frozen manifests • compiler 744a06a912da / pipeline bbe44c1151ac • no application speedup or target qualification claimed